The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5

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We construct a discrete graph encoding of the three-dimensional isotropic harmonic oscillator on the holomorphic sector of S^5 and verify by exact rational arithmetic that the resulting graph and its projection to physical energies contain no transcendental constants. The Bargmann–Segal transform [7, 9] maps the harmonic oscillator Hamiltonian to $\hbar\omega(\hat{N} + 3/2)$ on the Fock–Bargmann space, where $\hat{N}$ is the Euler number operator on holomorphic monomials. Restricting to the unit sphere $S^5 \subset \mathbb{C}^3$ identifies the eigenspaces with the symmetric SU(3) irreps $(N, 0)$, of dimension $(N + 1)(N + 2)/2$, which match the harmonic oscillator shell degeneracies exactly. The graph constructed from these states with dipole transition weights computed by exact `fractions.Fraction` arithmetic is bit-exactly π-free at $N_{\max} = 5$ (56 nodes, 165 edges, zero irrationals encountered). The only π in the construction is the Gaussian normalization π^{-3} of the continuous Bargmann measure, which the discrete graph never invokes. The one-mode version of that measure is the large-spin limit of a spin's coherent-state measure on S^2, so the oscillator is also reached from the compact label set of a spin, with the π entering only in the limit (Section III C). The Coulomb problem on S^3 and the harmonic oscillator on S^5 are both natural sphere quotients, but they live on different spheres with different geometric character (Riemannian vs complex), have natural operators of different order (second vs first), and yield spectra of different growth (quadratic vs linear). These differences do not decide where π enters. In the Kustaanheimo–Stiefel frame the Coulomb problem is a four-dimensional oscillator cut by one constraint: its label set, its n^2 degeneracy and the Casimir $n^2 - 1$ sit exactly on the oscillator's integer lattice, the projection is linear at fixed energy (the charges of the Coulomb Sturmians, $Z = np_0$), and the lift costs one fibre volume, $\pi/4$, which cancels in every ratio. The oscillator's spectral zeta values carry π^2 at even argument, as those of S^3 do. So π enters both sides in the same two ways, through a measure and through spectral sums (Section IV B), and what separates the two problems is representation theory and arithmetic.

I. INTRODUCTION

Fock's 1935 stereographic projection [1] turns the momentum-space hydrogen Schrödinger equation into a rotation-invariant integral equation on the three-sphere S^3. Hydrogen's bound states are the S^3 harmonics — the eigenfunctions of Δ_{S^3} — and on them the bound-state Hamiltonian is $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$, i.e. $E_n = -Z^2/(2(C+1))$ with $C = n^2 - 1$. The kernel $1/|\omega - \omega'|^2$ is zonal and proportional to $(1 - \Delta_{S^3})^{-1/2}$, with eigenvalue $2\pi^2/n$ on the degree-$(n-1)$ harmonics; that eigenvalue fixes $p_0 = Z/n$. The kernel is not the Green's function of Δ_{S^3}. Paper 0 [21] built a lattice on this construction: nodes labeled by quantum numbers (n, l, m)—exactly the label set of the S^3 harmonics, on which the Laplace–Beltrami operator acts as the diagonal $SO(4)$ Casimir $n^2 - 1$—and edges from angular-momentum selection rules. The lattice's graph Laplacian does *not* converge to the continuous Laplace–Beltrami operator (it preserves l and splits into $n_{\max}$ disconnected blocks, Paper 0 §VI; corrected 2026-09-25). The Rydberg energies $E_n = -Z^2/(2n^2)$ follow from the continuum eigenvalues $n^2 - 1$ through Fock's energy shell $p_0^2 = -2E$, a nonlinear map; the matched kinetic constant $\kappa = -1/16$ does not map $n^2 - 1$ onto them (it maps the graph's extremal eigenvalue $\lambda_{\max} \to 8$ onto the ground state, an Observation, Paper 7). Paper 7 [23] validated the continuum S^3 chain with eighteen symbolic checks. Paper 18 [25] identified a transcendental cost in the projection: π enters through the spectral zeta values of S^3 as a *calibration* exchange constant.

The harmonic oscillator $V(r) = \frac{1}{2}m\omega^2 r^2$ has a different dynamical symmetry group (SU(3) instead of SO(4)) and a qualitatively different spectrum (linear in shell index instead of inverse-square). The natural question is whether an analogous discretization exists for the harmonic oscillator, and if so, what role π plays.

We answer both questions in this paper. The discretization exists, via the Bargmann–Segal transform [7, 9] applied to the Hardy space $H^2(S^5)$. The graph is bit-exactly π-free at every finite $N_{\max}$. The single π in the entire construction appears in the Gaussian normalization π^{-3} of the continuous Bargmann measure, which is never used by the discrete graph. We further catalogue the Coulomb / harmonic oscillator asymmetry and test where π enters each side. In the Kustaanheimo–Stiefel frame the Coulomb problem is itself a constrained oscillator, so the type of projection is set by which number is held fixed, not by the geometry, and the spectral zeta values of both sides carry π. The placement of π therefore does not separate the two problems; the layers that do are representation-theoretic and arithmetic (Section IV D). The construction also bears on the framing of the fine-structure constant Observation [22] and on the precision of nuclear shell-model qubit Hamiltonians built on harmonic oscillator single-particle bases [27].

The plan of the paper is as follows. Section II summarizes the Bargmann–Segal construction and its identification with the SU(3) symmetric representation theory. Section III constructs the graph and states the π-free certificate (Theorem 1); Section III C reaches the one-mode Bargmann space from the label set of a spin by the large-spin contraction. Section IV catalogues the Coulomb / HO structural asymmetry and tests, in the Kustaanheimo–Stiefel frame, where π enters (Proposition 1). Section V states the harmonic oscillator rigidity theorem (Theorem 2), the dual of the Fock projection rigidity theorem of Paper 23 [27]. Section VI discusses implications for the α Observation, for nuclear quantum simulation, and for the framework's universal / Coulomb-specific partition.

II. THE BARGMANN–SEGAL CONSTRUCTION

A. The Bargmann transform

The Bargmann transform [7] is the integral operator

$$[Bf](z) = \pi^{-3/4} \int_{\mathbb{R}^3} \exp\Big(-\tfrac{1}{2}(z^2+x^2)+\sqrt{2}\, z\cdot x\Big)\, f(x)\, d^3x, \tag{1}$$

mapping $L^2(\mathbb{R}^3)$ unitarily onto the Fock–Bargmann space $F^2(\mathbb{C}^3)$ of holomorphic functions on $\mathbb{C}^3$ that are square-integrable against the Gaussian measure

$$d\mu(z) = \pi^{-3}\, e^{-|z|^2}\, d^6z. \tag{2}$$

The harmonic-oscillator eigenstates $|N,l,m\rangle$ map under B to solid harmonic polynomials of total degree N in the complex coordinates z_1, z_2, z_3. The transform is unitary, so the eigenvalue equations carry across exactly.

B. The Euler operator

The number (Euler) operator on $F^2(\mathbb{C}^3)$ is

$$\hat{N} = \sum_{i=1}^{3} z_i \frac{\partial}{\partial z_i}, \tag{3}$$

which acts on the monomial $z_1^a z_2^b z_3^c$ with eigenvalue $a+b+c = N$. By multinomial counting, the eigenspace $\hat{N} = N$ has dimension

$$\dim\big\{z_1^a z_2^b z_3^c \,:\, a{+}b{+}c = N\big\} \;=\; \binom{N+2}{2} \;=\; \frac{(N+1)(N+2)}{2} \tag{4}$$

which matches the 3D harmonic oscillator shell degeneracy and the Weyl dimension formula for the symmetric irreducible representation $(N,0)$ of SU(3).

Under the Bargmann transform, the harmonic oscillator Hamiltonian maps to

$$H_{\mathrm{HO}} \;=\; -\tfrac{1}{2m}\nabla^2 + \tfrac{1}{2}m\omega^2 r^2 \;\overset{B}{\longleftrightarrow}\; \hbar\omega\big(\hat{N}+\tfrac{3}{2}\big), \tag{5}$$

so the spectrum is

$$E_N \;=\; \hbar\omega\big(N+\tfrac{3}{2}\big) \tag{6}$$

on the eigenspace of dimension $(N+1)(N+2)/2$. The HO Hamiltonian is a linear affine function of the Euler operator; the projection from the discrete eigenvalue to the physical energy involves only the universal scale $\hbar\omega$ and the additive zero-point shift $3/2$.

The unitary equivalence of the Schrödinger and Bargmann pictures is rigorous and originates with Bargmann [7] and Segal [8]; the modern functional-analytic treatment is due to Hall [9], who extended the construction to arbitrary compact Lie groups.

C. The S^5 Hardy space and SU(3) symmetric representations

The unit sphere $S^5 = \{z \in \mathbb{C}^3 \,:\, |z| = 1\}$ carries a Cauchy–Riemann (CR) structure inherited from $\mathbb{C}^3$. The Hardy space $H^2(S^5)$ consists of boundary values of holomorphic functions on the unit ball $B^3 \subset \mathbb{C}^3$ that are square-integrable on S^5 with respect to the surface measure. Restriction to S^5 identifies $H^2(S^5)$ with the closure of the polynomial algebra $\mathbb{C}[z_1,z_2,z_3]$, with the natural SU(3) action by linear transformations of the coordinates.

The decomposition by polynomial degree gives an orthogonal direct sum

$$H^2(S^5) \;=\; \bigoplus_{N=0}^{\infty} \mathcal{H}_N, \qquad \dim \mathcal{H}_N = \frac{(N+1)(N+2)}{2}, \tag{7}$$

where $\mathcal{H}_N$ is the space of homogeneous degree-N holomorphic polynomials. Each $\mathcal{H}_N$ transforms as the SU(3) irreducible representation $(N,0)$ — the symmetric tensor representation of rank N on the fundamental **3**.

By the Borel–Weil theorem [12], this is the same data as the space of holomorphic sections of the line bundle $\mathcal{O}(N)$ over $\mathbb{CP}^2$, where $\mathbb{CP}^2 = \mathrm{SU}(3)/\mathrm{U}(2)$ is the natural quotient. The fibration

$$S^1 \;\longrightarrow\; S^5 \;\longrightarrow\; \mathbb{CP}^2 \tag{8}$$

is the structural analog of the Hopf fibration $S^1 \to S^3 \to S^2$ that played a central role in Paper 7 [23] and the α Observation of Paper 2 [22].

The CR Laplacian (Kohn Laplacian) $\Box_b$ on the holomorphic sector of S^5, the Dolbeault Laplacian on $\mathcal{O}(N) \to \mathbb{CP}^2$, and the Euler operator on the Bargmann monomials all act on the same SU(3) irreducibles $(N,0)$ and produce spectra that are linear in the shell index N up to additive and multiplicative normalizations [10]. A genuinely spherical harmonic-oscillator analogue — one whose configuration space is the sphere itself and which, unlike the flat-space oscillator, does not factor into one-dimensional modes — has been studied independently in

recent mathematical work [35], which identifies the two-dimensional case as critical. That construction is distinct from the flat-space isotropic oscillator discretized here via the S^5 Hardy sector; we note it only as an independent recent appearance of harmonic-oscillator dynamics with sphere symmetry. For the discretization in Section III we use the Bargmann description, which is the most elementary: the Euler operator is diagonal on the monomial basis, requires no differential geometry, and is amenable to immediate exact arithmetic.

III. THE BARGMANN GRAPH

A. Graph construction

a. Nodes. The nodes of the Bargmann graph at cutoff $N_{\max}$ are triples (N, l, m_l) with $N = 0, 1, 2, \ldots, N_{\max}$; for each N, the orbital angular momentum runs over $l = N, N-2, \ldots, 1$ or 0 (same parity as N); and for each l, the magnetic quantum number runs over $m_l = -l, \ldots, +l$. This is the SU(3) $\to$ SO(3) branching of the symmetric irrep $(N, 0)$. The total node count is

$$|V(N_{\max})| \;=\; \sum_{N=0}^{N_{\max}} \frac{(N+1)(N+2)}{2}. \tag{9}$$

At $N_{\max} = 5$ this gives $1 + 3 + 6 + 10 + 15 + 21 = 56$ nodes.

b. Edges. Edges connect states under SU(3) dipole selection rules:

$$\Delta N = \pm 1, \qquad \Delta l = \pm 1, \qquad \Delta m_l = 0, \pm 1. \tag{10}$$

These are the matrix elements of the position operator z_q ($q = 0, \pm 1$ in spherical components) between adjacent shells, which raise N by one and change l by ± 1 by the standard angular-momentum dipole rule.

c. Edge weights. For each allowed transition the edge weight is the squared matrix element

$$w_{(N,l,m_l)\to(N+1,l',m_l')} \;=\; R^2_{n_r,\, l\to l'} \;\times\; \begin{pmatrix} l & 1 & l' \\ m_l & q & -m_l' \end{pmatrix}^2, \tag{11}$$

where the radial reduced matrix elements (in HO units $\hbar = m = \omega = 1$) are

$$\big|\langle n_r, l+1 \,|\, r \,|\, n_r, l\rangle\big|^2 = n_r + l + \tfrac{3}{2} = \tfrac{2n_r+2l+3}{2}, \tag{12}$$

$$\big|\langle n_r+1, l-1 \,|\, r \,|\, n_r, l\rangle\big|^2 = n_r + 1, \tag{13}$$

and the squared rank-1 Wigner $3j$ coefficients factorize into elementary rationals via the closed-form expressions of Edmonds [16] and Varshalovich [17]. Both factors in Eq. (11) are exact rational numbers, so the edge weight is exact rational.

d. Implementation. The reference implementation in `geovac/nuclear/bargmann_graph.py` uses Python's `fractions.Fraction` type throughout the construction. No floating-point arithmetic is used in the build path; the squared $3j$ tables are evaluated symbolically from the closed-form expressions, and the radial reduced matrix elements are constructed directly from the integer formulas above. Conversion to floating point occurs only when the dense Hamiltonian or adjacency is required for numerical eigensolvers.

B. The π-free certificate

Theorem 1 (Bargmann graph rationality)**.** *Let $G(N_{\max})$ be the Bargmann graph constructed in Section III with diagonal entries*

$$D_v \;=\; \mathrm{Fraction}(2N_v + 3,\, 2)\,, \tag{14}$$

in units of $\hbar\omega$, and adjacency weights given by Eq. (11) *computed in exact rational arithmetic. Then every diagonal entry and every nonzero adjacency weight is a rational number, and no transcendental constant appears in the construction at any finite $N_{\max}$.*

Proof. The diagonal entries are Fraction$(2N + 3, 2)$ by construction (Section II B, Eq. (6)). The adjacency weights are products of two rational factors: the squared radial reduced matrix elements (which are rational linear functions of the integer quantum numbers n_r and l) and the squared rank-1 Wigner $3j$ coefficients (which are rational quotients of polynomials in the integer quantum numbers l, l', m_l, m_l' by the Edmonds [16] formulas). The product of two rationals is rational, and the construction terminates after finitely many additions and multiplications of rationals at any finite $N_{\max}$. □ □

a. Numerical verification. An independent run of the function `verify_pi_free(N_max=5)` in the reference implementation returns the certificate:

```
N_max:  5
n_nodes:  56
n_edges:  165
all_diagonal_rational:  True
all_adjacency_rational:  True
irrationals_encountered:  []
pi_free:  True
```

The 17 supporting unit tests are listed in `tests/test_bargmann_graph.py` and cover node enumeration, shell degeneracies, parity of l within each shell, the diagonal HO spectrum, the dipole selection rules, Hermiticity of the adjacency, exact rationality of every entry, and graph Laplacian positivity. All 17 pass.

b. Where the only π lives. The single π in the construction appears in the Gaussian normalization π^{-3} of the continuous Bargmann measure Eq. (2). In Paper 18's exchange constant taxonomy [25], this is an *embedding* exchange constant: it quantifies the cost of representing

the Bargmann Hilbert space as a measure space in 6 real dimensions, and it is fixed by the requirement that the Gaussian have unit total integral. The discrete graph never invokes this normalization. No analogous π enters the spectrum, the degeneracies, the selection rules, the Wigner $3j$ coefficients, or the radial matrix elements. Spectral sums over the spectrum do carry π at even argument, as on S^3 (Section IV B).

C. A spin-side entrance: the large-spin contraction

A spin S has the finite label set $m = -S, \ldots, S$. As S grows, the neighbourhood of the pole $m = S$ flattens and the spin algebra contracts to the one-mode oscillator algebra [2, 3], so the one-mode Bargmann space is reached from a compact label set. **[PRIOR ART]** The mathematics is classical; what is recorded here is the accounting of what the release leaves standing and what it lets in.

With $n = S - m$ and $A_S^\dagger = S_-/\sqrt{2S}$,

$$\langle n{+}1|A_S^\dagger|n\rangle = \sqrt{n+1}\,\sqrt{1-\frac{n}{2S}}, \qquad [A_S, A_S^\dagger]\,|n\rangle = \Big(1{-}\frac{n}{S}\Big)|n\rangle \tag{15}$$

[SYMBOLIC PROOF] The label n survives as the oscillator quantum number, and the only thing released is the cap $n \le 2S$. The limit is not uniform: it fails for n of order S.

The operator $X_S = S_x/\sqrt{S}$ has eigenvalues $x = m/\sqrt{S}$, spaced $1/\sqrt{S}$, so its spectrum becomes dense on the whole real line. In the pole state the probability of eigenvalue x, per unit x, is

$$p_S(x) = \sqrt{S}\binom{2S}{S+m}4^{-S} = \frac{e^{-x^2}}{\sqrt{\pi}}\Big[1{+}\frac{c_1(x)}{S}{+}\frac{c_2(x)}{S^2}{+}\cdots\Big], \tag{16}$$

with $c_1 = -\frac{1}{8} + \frac{x^2}{2} - \frac{x^4}{6}$, $c_2 = \frac{1}{128} - \frac{11x^2}{48} + \frac{19x^4}{48} - \frac{3x^6}{20} + \frac{x^8}{72}$, and no half-integer powers of $1/S$; the series is asymptotic, not convergent. **[MEASURED]** At $S = N^2$ the density is an exact rational; a fit over $N = 40, \ldots, 400$ gives the limit to 10^{-35} and c_1 to 10^{-29} at seven values of x, and c_2 is one polynomial fitting the same seven points to 10^{-26}. The eigenvectors follow the same pattern: $S^{1/4}\langle S, S-n|X_S = x\rangle$ tends to the Hermite function $\psi_n(x)$ at rate $1/S$, for $n \le 4$ (**[MEASURED]**; the Krawtchouk-to-Hermite limit of the Askey scheme [6]).

a. Where the π enters. At every finite S the label side carries no π: the matrix elements are square roots of rationals, the eigenvector components of S_x have rational squares, and $p_S(x)/p_S(0)$ is rational. The constants $\pi^{1/2}$ and e^{-x^2} appear only as limits of rational label data, $\pi^{-1/2}$ through Wallis's limit of $\sqrt{S}\binom{2S}{S}4^{-S}$. On the continuous side, the spin coherent states resolve the identity with the measure $(2S+1)/(4\pi)\,d\Omega$ on the sphere [4, 5]. Under $z = w/\sqrt{2S}$ in the stereographic coordinate this measure, times the coherent-state weight, contracts to $\pi^{-1}e^{-|w|^2}\,d^2w$, the one-mode form of Eq. (2).

[SYMBOLIC PROOF] The Bargmann π is therefore the large-S limit of the sphere's $1/(4\pi)$, the normalization by $\mathrm{Vol}(S^2)$. Paper 18 reads a π of this kind as its volume slot M1, as a reading and not a derivation [25], and this one takes the same tag. Only one contraction is covered here (near a pole); the non-compact real form SU(1, 1) and Wigner's continuous-spin representations are other routes and are not tested. Guard: `tests/test_paper24_large_spin_door.py`.

D. Hamiltonian and graph Laplacian roles

The HO Hamiltonian on the Bargmann graph is a diagonal matrix:

$$H_{\mathrm{HO}} \;=\; \hbar\omega\,\mathrm{diag}\big(N_v + \tfrac{3}{2}\big)_{v\in V}\,. \tag{17}$$

The eigenvalues are $\hbar\omega(3/2, 5/2, 7/2, 9/2, \ldots)$ with multiplicities $1, 3, 6, 10, \ldots = (N{+}1)(N{+}2)/2$. No off-diagonal terms are needed to recover the spectrum: it is simply the diagonal.

The adjacency matrix A encodes dipole transition amplitudes between HO states (electromagnetic absorption and emission lines). It plays no role in the construction of the spectrum; its role is to describe the connectivity structure that determines selection rules for transition processes.

This is the *same* situation as in the Coulomb / S^3 case of Paper 0 [21] and Paper 7 [23], contrary to an earlier version of this section. There too the exact spectrum is a diagonal: on the (n, l, m) labels the S^3 Laplace–Beltrami operator is the $SO(4)$ Casimir $\mathrm{diag}(n^2{-}1)$, and the graph Laplacian $D{-}A$ does *not* converge to it (it preserves l and splits into $n_{\max}$ disconnected blocks; Paper 7). In both graphs, therefore, the off-diagonal structure is spectroscopically informative but spectrally inert. (The version printed until 2026-09-25 called this "the first key structural difference" between the two discretizations, on the premise that $D - A$ converges to Δ_{S^3}; that premise is false, and the layer is withdrawn — see the note at the head of Section IV.) The asymmetries that survive are analysed in Section IV.

IV. THE COULOMB / HO ASYMMETRY

The Bargmann graph and the Coulomb S^3 graph are both natural discretizations of central-potential quantum mechanics, but they differ in every structural feature except the use of angular momentum as a labeling system. Table I summarizes the comparison.

Note on the layer count. The Coulomb/HO asymmetry has been refined to seven layers catalogued, five standing (Layer 1 withdrawn 2026-09-25, Layer 2 withdrawn 2026-10-19) (enumerated in §IV D) across Sprint 5 Track S5 (gauge-content), Paper 30 (SU(2) Wilson on

S^3), Sprint ST-SU3 (SU(3) Wilson on S^5), Sprints L2-E/L2-F.1 (modular wedge), Paper 51 (gravity termination), Sprint Q5′-S5 (Mellin parity), and the Hodge-circle arithmetic computation (August 2026). *Layer (i) is withdrawn (2026-09-25):* it rested on $L_0 = D - A$ converging to Δ_{S^3}, which is false (Paper 7); on both spheres the spectrum is diagonal on the labels and L_0 is spectrally inert. *Layer (ii) is withdrawn (2026-10-19):* π enters both sides through a measure and through spectral sums at even argument (Section IV B). Five of the seven catalogued layers stand. The first three layers as catalogued are: (i) spectrum-computing role of L_0 [withdrawn]; (ii) calibration π [withdrawn]; (iii) Wilson gauge with natural matter coupling — with the universal Wilson–Hodge combinatorial vocabulary as the shared (**universal**) residue, not itself one of the seven. Sprint ST-SU3 split the Wilson layer into "gauge alone" (universal) versus "gauge with natural matter coupling" (Coulomb-specific): on S^3, $S^3 = \mathrm{SU}(2)$ means the matter spinor lives in the same representation as the Wilson links, so gauge-matter coupling is argued to be natural (not yet realised as a coupled construction, Paper 30); on S^5, pure-gauge SU(3) is well defined, but the matter representations $(N, 0)$ have dimensions that grow with N, so a uniform link-rep matter coupling is not available, and any coupling reverts to a Clebsch–Gordan intertwiner — the same obstruction Sprint 5 identified. See `debug/st_su3_wilson_memo.md` § 5.4.

A. The causal chain

The first three steps below follow from one another; the fourth, the projection type, holds in the energy frame and depends on which number is held fixed (Section IV B).

1. *Geometry choice.* SO(4) symmetry of the Coulomb problem (the Runge–Lenz vector closes with the angular momenta into $\mathfrak{so}(4)$) makes S^3 the natural manifold, by Fock's reduction [1]. SU(3) symmetry of the harmonic oscillator (the ladder operators $a_i = (x_i + ip_i/m\omega)\sqrt{m\omega/2\hbar}$ transform as the fundamental **3** and generate $\mathfrak{su}(3)$ under commutation) makes $\mathbb{C}^3$ and its quotients $\mathbb{CP}^2$ and S^5 natural, by [11].

2. *Operator order.* A real Riemannian metric on S^3 makes the scalar Laplace–Beltrami operator (second-order in derivatives) the natural object. A complex Cauchy–Riemann or Kähler structure on S^5 or $\mathbb{CP}^2$ makes the Dolbeault, Kohn, or Euler operators (first-order in $\hat{N}$ on holomorphic sections) natural. This is not a free choice; it follows from the requirement that the operator be invariant under the dynamical symmetry group acting holomorphically on the manifold's structure.

3. *Spectrum growth.* A second-order self-adjoint operator on a compact Riemannian manifold has eigenvalues that grow polynomially with shell index, typically as n^2 for round spheres (Weyl asymptotics). A first-order equivariant operator on the symmetric irreducible representations of a compact Lie group has eigenvalues linear in the highest weight. These growths are different by definition.

4. *Projection type.* Mapping a quadratic mathematical eigenvalue $\lambda_n = n^2 - 1$ to an inverse-square physical energy $E_n = -Z^2/(2n^2)$ is a *nonlinear* operation: $E = -Z^2/[2(\lambda + 1)]$. Mapping a linear mathematical eigenvalue $\lambda_N = N$ to a linear physical energy $E_N = \hbar\omega(N + 3/2)$ is a *linear-affine* operation.

The chain stops at step 4. A further step, that a nonlinear projection needs spectral-zeta regularization which brings π while a linear-affine one brings none, fails on two counts, both shown in Section IV B: the projection type of step 4 depends on which number is held fixed, and the oscillator's spectral zeta values carry π^2 at even argument as those of S^3 do.

Proposition 1 (π does not separate the two columns)**.** *(i) In the Kustaanheimo–Stiefel frame the hydrogen label set, its degeneracy n^2 and the Casimir n^2-1 hold exactly on the integer lattice of the four-dimensional isotropic oscillator restricted to the kernel of the fibre generator, and at fixed p_0 the Coulomb projection is linear, $Z_n = n\,p_0$. (ii) The spectral zeta values of the oscillator ladders carry π^2 at even integer argument, as those of the S^3 Casimir ladder do. Hence neither the projection type nor the presence of π in spectral sums distinguishes the Coulomb problem from the three-dimensional isotropic harmonic oscillator.*

Part (i) is classical: the lift is Kustaanheimo and Stiefel's [18], the exchange of energy and charge is coupling-constant metamorphosis [19], and the linear charges are those of the Coulomb Sturmians [20]. Part (ii) is elementary. Both are checked in exact arithmetic in Section IV B.

B. The Kustaanheimo–Stiefel frame

Lift $\mathbb{R}^3$ to $\mathbb{R}^4 = \mathbb{C}^2$ by the Hopf map, $x_1 + ix_2 = 2z_1\bar{z}_2$, $x_3 = |z_1|^2 - |z_2|^2$, $r = |z|^2$ [18]. At fixed energy $E = -p_0^2/2$ the Coulomb equation becomes that of the four-dimensional isotropic oscillator $H = p_u^2/8 + 2\omega^2|u|^2$ with $\omega = p_0/2$, restricted to the states annihilated by the generator of the fibre $z \mapsto e^{i\phi}z$. Energy and coupling exchange roles [19]: the energy is consumed as the frequency and the charge comes out as the eigenvalue. We work with Bargmann monomials in the circular variables $(A_\pm, B_\pm)$ of the planes (u_1, u_2) and (u_3, u_4), with level N the total degree and fibre charge $Q = (a_+ - a_-) + (b_+ - b_-)$; every generator used has integer matrix entries on the monomials (halves for the z-

TABLE I. Structural comparison of the Coulomb and harmonic oscillator discretizations. Each row is a feature; each column is a system. Rows 2–4 follow from one another: the geometry fixes the operator order, which fixes the spectrum growth. Rows 6 and 7 describe the energy frame; at fixed energy the Coulomb projection is linear too (Section IV B). Both columns carry π in a measure and in spectral sums.

Property	Coulomb (S^3, Riemannian)	HO (S^5 Hardy, complex)
Symmetry group	SO(4)	SU(3)
Manifold	S^3	S^5 (holomorphic sector)
Natural operator	Laplace–Beltrami (2nd order)	Euler / Dolbeault (1st order)
Eigenvalue λ_N	n^2-1 (quadratic in n)	N (linear in N)
Shell degeneracy	n^2	$(N+1)(N+2)/2$
Projection $E(\lambda)$, energy frame	$E_n = -Z^2/[2(\lambda+1)]$ (nonlinear)	$E_N = \hbar\omega(\lambda+3/2)$ (linear-affine)
Projection constant	$p_0 = Z/n$ (at fixed p_0: $Z_n = np_0$, universal)	$\hbar\omega$ (universal)
Graph Laplacian role	encodes transitions only (spectrum = diagonal Casimir; $D-A \not\to \Delta_{S^3}$)	encodes transitions only
Where π enters	measure (Fock kernel $2\pi^2/n$), spectral sums	measure (π^{-3}), spectral sums
π-free graph	yes	yes

components). **[SYMBOLIC PROOF]** In exact integer and rational arithmetic:

1. *Labels.* The $Q = 0$ sector is empty at odd N (through $N = 27$) and has dimension n^2 at $N = 2(n-1)$ for $n \le 15$, holding $l = 0, \ldots, n-1$ once each under the diagonal SU(2) (by counting for $n \le 15$, by the minimal polynomial of L^2 for $n \le 8$).
2. *Casimir.* With $\mathrm{SU}(2)_+$ acting on (A_+, B_+) and $\mathrm{SU}(2)_-$ on (A_-, B_-) as the conjugate doublet, on each $Q = 0$ level $J_+^2 = J_-^2$ and $2(J_+^2 + J_-^2) = (n^2-1)\,\mathbf{1}$ for $n \le 10$: the diagonal SO(4) Casimir on the labels, with no Fock projection.
3. *Linear projection.* Each $Q = 0$ eigenfunction, pushed down to $\mathbb{R}^3$, satisfies the Coulomb Sturmian equation $(-\frac{1}{2}\nabla^2 - Z/r)\, f = -\frac{1}{2}p_0^2 f$ with $p_0 = 2\omega$ and $Z = \omega(N+2) = n\,p_0$ (all 14 states with $n \le 3$, symbolically; a wrong charge or exponent leaves a nonzero residual on states with radial nodes). The eigenvalue is linear in the label with one universal constant p_0.
4. *The price.* For two $Q = 0$ states the $\mathbb{R}^4$ inner product equals $(\pi/4)\int f f'\, d^3x/r$, the same constant for each of the eight nonzero pairs checked ($n \le 3$) and free of ω: the fibre volume. The lift brings one π, which cancels in every normalized quantity.
5. *Spectral sums.* At even s both sides carry π. The S^3 Casimir ladder gives $\sum_{n\ge 2} n^2(n^2-1)^{-2} = \pi^2/12 + 1/16$ and $\sum_{n\ge 2} n^2(n^2-1)^{-4} = 5/256 - \pi^2/96 + \pi^4/720$; the four-dimensional ladder gives $\sum_n n^2\, n^{-s} = \zeta(s-2)$. With $g_N = (N+1)(N+2)/2$ the oscillator Euler ladder gives $\sum_{N\ge 1} g_N N^{-s} = \frac{1}{2}[\zeta(s-2) + 3\zeta(s-1) + 2\zeta(s)]$, which at $s = 4$ also carries $\zeta(3)$, and the energy ladder gives $\sum_{N\ge 0} g_N (N+\frac{3}{2})^{-4} = \pi^2/4 - \pi^4/48$.

So rows 6 and 7 of Table I record which number is held fixed, and the π of spectral sums is common to both columns. The $\pi/4$ is read as a volume (M_1) slot, at Paper 18's tier. The continuum, which maps to an inverted oscillator, and two-centre regularizations are not tested. Pre-registered predictions and their scoring: `docs/predictions/2026-10-19_ks_lift.md`; guard `tests/test_paper24_ks_lift.py`.

C. The exchange constant taxonomy on both sides

Paper 18 [25] classifies exchange constants into six tiers, beside the algebraic (no-exchange) case of content that stays on the discrete labels: *intrinsic* (Weyl constants $\pi^{d/2}$ fixed by a manifold's dimension and volume alone), *conformal/calibration* (from the conformal projection of the compact manifold to flat space—for S^3, Fock's energy shell), *embedding* (single seeds from a spectral basis meeting a coordinate singularity of the continuous space used for measurement), *algebraic-implicit* (parameter-dependent functions with inherited transcendental content), *composition* (the cost of factorizing into composed coordinates), and *inner-factor input data* (calibration data of the almost-commutative inner factor, supplied externally).

The Coulomb S^3 side carries calibration π (Paper 2's $K = \pi(B + F - \Delta)$ [22]) and embedding π (the electron-electron $1/r_{12}$ Slater integrals and the radial normalization of hydrogenic functions). The oscillator side carries the embedding π^{-3} of the Gaussian Bargmann measure, which the graph never uses, and π^2 in its spectral zeta values (Section IV B), the same kind of spectral-sum π as the S^3 side's. The calibration tier is therefore not tied to a nonlinear projection.

D. Fourth layer: wedge-generator Pythagorean orthogonality

The note at the head of Section IV records the original three layers of the Coulomb/HO asymmetry: the spectrum-computing role of L_0 (Coulomb-specific as catalogued; withdrawn 2026-09-25), the calibration π (Coulomb-specific as catalogued; withdrawn 2026-10-19), and Wilson gauge with natural matter coupling (Coulomb-specific). Sprints L2-E + L2-F.1 (May 2026) identified a fourth layer at the level of the (posited) wedge generator:

- Layer 1: spectrum-computing role of L_0 [retracted 2026-09-26: coulomb-ho-layer1-and-three-discriminators] (*withdrawn* 2026-09-25: L_0 is spectrally inert on S^3 too, since it does not converge to Δ_{S^3}; Paper 7).
- Layer 2: calibration π (*withdrawn* 2026-10-19: π enters both sides through a measure and through spectral sums at even argument, and the projection type depends on which number is held fixed; Section IV B).
- Layer 3: Wilson lattice gauge with natural matter coupling (Coulomb-specific, via Papers 25, 30; S^5 is gauge-only per Sprint ST-SU3).
- Layer 4: Pythagorean orthogonality of the posited wedge generator and the wedge Dirac on the hemispheric wedge (its Bisognano–Wichmann identification is open, Paper 42; Coulomb-specific [ARGUED]: on the S^5 Hardy sector the natural analog operands are not orthogonal, a comparison of chosen analog operators rather than a theorem).
- Layer 5: spectral-action gravity termination (Paper 51). The CH Dirac spectral action on S^{2m+1} terminates at exactly $(m{+}1)$ power-law terms; only S^3 ($m = 1$) gives *pure Einstein gravity* (2 terms: cosmological + EH, no R^2 or higher-curvature corrections). S^5 ($m = 2$) terminates at 3 terms (includes R^2 Gauss–Bonnet). The mechanism is the Bernoulli identity for odd polynomials at half-integer shifts.
- Layer 6 *(new)*: master Mellin engine chirality-parity selection rule (Sprint Q5'-S5-falsifier, June 2026; `debug/sprint_q5p_s5_falsifier_memo.md`). On the Camporesi–Higuchi S^3 spectral triple at every finite $n_{\max}$, the Mellin source $\mathrm{Tr}(D^k\, e^{-tD^2})$ exhibits a bit-exact $\mathbb{Z}/2$ chirality-parity factorization (Sprint Q5'-CH-1): plain trace vanishes identically at odd k, supertrace vanishes identically at even k. On the Bargmann–Segal Hardy sector at every finite $N_{\max}$ this rule fails on both load-bearing structural conditions: the Hardy spectrum is strictly positive (no $\pm$-partner) and no half-integer-weight chirality grading exists on the $(N, 0)$ tower (the *angular* weight; the zero-point $\frac{3}{2}$ is a different, metaplectic half-integer — see the Layer-7 companion note below). This propagates the Coulomb/HO asymmetry from the wedge-generator level (Layer 4) and the spectral-action level (Layer 5) to the master Mellin engine level, naming a CH-substrate-specific structural prerequisite for the cosmic-Galois U^* enrichment program (Paper 55 § subsec:open_m2_m3).
- Layer 7 *(new)*: CM/Hodge-circle arithmetic of the compact flow (August 2026; the flow-level identification is Observation-tier in Paper 56, while the S^5 obstructions below are exact; `tests/test_paper24_hardy_circle_arithmetic.py`). On S^3 the compact circle generated by the posited $2J_z$ wedge generator (its Bisognano–Wichmann identification is open, Paper 42) has $\pm$-symmetric integer generator spectrum, so its $\beta/4$ quarter-period point *is* a $\mathbb{Q}$-rational complex structure J, $J^2 = -\mathbf{1}$, making the fundamental doublet a polarized weight-one $\mathbb{Q}$-Hodge structure with CM by $\mathbb{Q}(i)$ (Paper 56). The S^5 Hardy phase circle — the $U(1)$ generated by the Euler operator — is equally compact and even carries a *faithful* μ_4 (the metaplectic zero-point $\frac{3}{2}$ lifts the graph's bipartition parity $(-1)^N$ to order four), yet admits *no* CM Hodge structure, for three independent exact reasons: (i) the generator $N + \frac{3}{2}$ is strictly positive, while $\det(\cos t\,\mathbf{1} + \sin t\, J) = 1$ forces a traceless (i.e. $\pm$-paired) generator on any Hodge 2-plane; (ii) on the substrate's own $\mathbb{Q}$-form (the π-free (N, l, m) basis) the quarter point acts by i^N (projectively; the metaplectic $\frac{3}{2}$ contributes a global phase $e^{3\pi i/4}$), $\mathbb{Q}$-rational exactly where it squares to $+\mathbf{1}$; (iii) a weight-one structure needs the conjugate half $(0, N)$ — precisely the antiholomorphic sector the Hardy construction discards, whereas the Dirac's chirality-doubled $\pm(n + \frac{3}{2})$ spectrum carries it for free. Layer 7 is a new consequence level (arithmetic/Tannakian: the field of definition of the flow's Hodge group), standing to Layers 4–6 as those stand to Layers 1–3.

A companion *non*-layer belongs beside Layer 7, because it is the half of the S^3 arithmetic that does *not* separate the two sides: conductor-4 *period* content transfers. The Bargmann graph is bipartite (every edge has $|\Delta N| = 1$), so $(-1)^N$ is a substrate-natural $\mathbb{Z}_2$ grading, and the graded Hardy spectral zeta evaluates in closed form to $Z_-(s) = \sum_N (-1)^N g_N\,(N + \frac{3}{2})^{-s} = 2^{s-3}(\beta(s) - \beta(s-2))$ — exactly one quarter of the Dirac-tower value of Paper 28's χ_{-4} theorem, because both towers carry the same "index + $d/2$" eigenvalues at $d = 3$ and their degeneracies differ by 4 (at $s = 4$: $Z_-(4) = 2(\beta(4) - G)$ with G Catalan's constant). The χ_{-4} is produced by the half-integer index shift times the $\mathbb{Z}_2$ grading — spin on S^3, the *metaplectic* zero-point on S^5, no spin required — while the ungraded Hardy

partition function $\sum_N g_N q^N = 1/(1-q)^3$ (the Hilbert series of $\mathbb{C}[z_1, z_2, z_3]$) is pure Tate, conductor 1. So the sharp split is: an order-2 datum lifted to order 4 suffices for conductor-4 *periods* on either side; the CM *Hodge structure* of Layer 7 additionally needs the ±-paired $\mathbb{Q}$-rational carrier, which only the spinor side has. (Honest cap: on the Dirac side the β-values are measured in a computed observable, while Z_- is not yet attached to any HO/nuclear observable — the identity is exact, its physical significance an observation. And the HO shift can be normal-ordered away, killing the grading's χ_{-4}, whereas the Dirac shift is chirality-forced — the residual asymmetry *inside* the period statement.)

The fourth-layer statement is structural rather than numerical. On the truncated Camporesi–Higuchi spectral triple at signature $(3,1)$ (the Lorentzian Krein-space lift of the S^3 Coulomb discretisation; Paper 42 § 11 [33], formerly Paper 43), the local Hamiltonian $H_{\text{local}} := K_L^{\alpha,W}/\beta$ built from the *posited* integer-spectrum wedge generator (its identification with the Bisognano–Wichmann boost is open, Paper 42) and the wedge-restricted Lorentzian Dirac D_W^L are Hilbert–Schmidt orthogonal (a chirality-parity identity):

$$\langle H_{\text{local}}, D_W^L \rangle_{\text{HS}} \;=\; 0$$

bit-exactly at every tested $n_{\max}$ and N_t, and independently of β (so across all six labelled witnesses, which are the same generator with κ_g relabelled; Paper 42 Theorem 11.14). This chirality-parity identity holds because H_{local} is diagonal in the unfolded wedge spinor basis (eigenvalues two_m$_j > 0$, the rotation generator around the Hopf axis) and independent of the chirality $\chi = \pm 1$, while D_W^L splits into a temporal part $i\gamma^0 \otimes \partial_t$ that is odd under the wedge chirality grading and a spatial part that is chirality-*diagonal* and sign-paired (eigenvalues $\chi\,(n_{\text{fock}} + \frac{1}{2})$), so the $\chi = \pm 1$ contributions of the spatial part cancel pair by pair in the trace, while the temporal part $i\gamma^0 \otimes \partial_t$ contributes nothing because it is chirality-odd. Both operators are diagonal in the spatial labels; what the identity uses is the sign pairing of the spatial part and the chirality-oddness of the temporal part, not a diagonal/off-diagonal split. The closed-form residual, which holds at $N_t = 1$ (Paper 42 Corollary 11.15; the orthogonality itself holds at every N_t), carries its only transcendental through the posited normalisation: the $1/\pi^2$ is $1/\beta^2$ at $\beta = 2\pi$, the square of the temporal-compactification π of Paper 35, not an output of the master Mellin engine:

$$\|H_{\text{local}} - D_W^L\|_F^2 \;=\; \kappa_g^2 \cdot \frac{S(n_{\max})}{4\pi^2} \;+\; D(n_{\max}),$$

with S and D pure rationals in $n_{\max}$ (cumulative wedge Casimir traces).

a. Why the fourth layer fails on the S^5 Hardy sector [ARGUED]. The orthogonality proof (Paper 42 Theorem 11.14) uses three ingredients: a chirality grading with ±-paired eigenspaces, a generator independent of that grading, and a first-order operator that is sign-paired under it ($D = \Pi\,|D|$). On S^3 the Camporesi–Higuchi spinor bundle supplies all three. On the Bargmann–Segal Hardy sector $H^2(S^5)$ the grading and the sign pairing are absent. Four named obstructions:

1. *No spinor sector on the $(N, 0)$ tower.* The symmetric irreducible representations of SU(3) are bosonic (integer m_l); the Hardy sector has no half-integer weight content, hence no spinor lift compatible with the Bargmann transform.

2. *No sign-paired first-order operator.* The HO's first-order operator is the Euler operator $\hat{N} + 3/2$ on the Hardy sector, which *is* the Hamiltonian, and it is strictly positive: there is no ±-partner for its eigenvalues and no grading to pair them (obstruction 1; Layer 6). Diagonality is *not* the discriminator — on S^3 both operands are diagonal too. The discriminator is the sign pairing: taking the natural analog operands, the BW-analog generator $K^{\alpha,W} = \text{diag}(2m_l > 0)$ and $\hat{N} + 3/2$, both are diagonal with strictly positive entries on the wedge, so $\langle K^{\alpha,W}, \hat{N} + \frac{3}{2} \rangle_{\text{HS}} = \sum_{\text{wedge}} 2m_l\,(N + \frac{3}{2}) > 0$ at every cutoff, where on S^3 the χ-paired terms cancel. The positivity is elementary; the choice of these two operands as the S^5 analog is [ARGUED] (the Hardy sector has no canonical wedge Dirac) and is not separately test-backed.

3. *No half-integer wedge.* The natural BW-analog generator $K^{\alpha,W} = \text{diag}(2m_l > 0)$ has *even*-integer spectrum, so the modular period closes at $\beta = \pi$ rather than $\beta = 2\pi$. The finite-cutoff 2π closure on S^3 is the automatic closure of the posited $2J_z$ generator with odd-integer spectrum (its Bisognano–Wichmann identification is open, Paper 42), so it is the half-integer m_j content that fixes that period; the continuum four-witness theorem at $\beta = 2\pi$ is prior art and is not what is being compared here.

4. *Coulomb/HO category mismatch resurfaces.* The layer-3 asymmetry already documented above propagates to the wedge-generator level: every structural ingredient Pythagorean orthogonality requires (spinor bundle with a chirality grading + a sign-paired first-order D; and, for the 2π period, a half-integer wedge) is the Coulomb-side outcome of layer 3 (layers 1 and 2 are withdrawn).

These four obstructions are the same blocker that prevents the cross-manifold tensor product $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\text{Hardy}(S^5)}$ at the AC-extension level (Paper 32 § VIII.C G4b), now showing up at the wedge-generator level as a sibling obstruction rather than a new one. The Coulomb/HO asymmetry (seven layers catalogued, five standing after the withdrawals of the spectrum-computing layer, 2026-09-25, and the calibration-π layer, 2026-10-19) is consistent across gauge-matter coupling, posited wedge-generator structure, spectral-action gravity termination,

master-Mellin chirality factorization, and Hodge-circle arithmetic.

b. Honest scope. The HS-orthogonality is empirically demonstrated, and it is a chirality-parity identity (an earlier version of this section explained it by a diagonal/off-diagonal subspace decomposition; withdrawn 2026-09-28, since the spatial wedge Dirac is chirality-diagonal and sign-paired). A formal proof at general $(n_{\max}, N_t)$ and every β is now given in Paper 42 Theorem 11.14 (§ 11.7): the orthogonality is an elementary chirality-parity identity about two chosen operators, independent of β, so the six labelled witnesses are one statement and it does not decide which generator a wedge state selects; the bit-exact panel at $n_{\max} \in \{1, \ldots, 5\}$ is consistent with it. Supporting memos: `debug/h_local_residual_pslq_memo.md`, `debug/six_witness_hs_orthogonality_memo.md`, `debug/pythagorean_extension_scoping_memo.md`.

V. THE HARMONIC OSCILLATOR RIGIDITY THEOREM

The Fock projection rigidity theorem of Paper 23 [27] states that the S^3 Fock projection $p_0 = \sqrt{-2E_n}$ maps a one-electron central-field Hamiltonian onto a rotation-invariant (SO(4)) integral equation on S^3 whose eigenfunctions are the S^3 harmonics (the eigenfunctions of Δ_{S^3}; the kernel is proportional to $(1 - \Delta_{S^3})^{-1/2}$, not the Green's function of Δ_{S^3}), if and only if the spectrum is l-degenerate within each n-shell, which is unique to the Coulomb potential by SO(4) symmetry.

The harmonic oscillator analog has the same structural form:

Theorem 2 (Bargmann–Segal rigidity for the 3D HO)**.** *The three-dimensional isotropic harmonic oscillator $V(r) = \frac{1}{2}m\omega^2 r^2$ is the unique central-field potential whose bound-state spectrum arises from the Euler operator on the Hardy space $H^2(S^5)$ restricted to the symmetric irreducible representations $(N, 0)$ of* SU(3)*.*

Proof. The argument has three independent classical inputs.

Step 1 (Jauch–Hill, 1940; uniqueness sharpening Wybourne / Iachello). Jauch and Hill [11] proved the forward direction: the three-dimensional isotropic harmonic oscillator possesses an SU(3) dynamical symmetry, constructing the explicit $\mathfrak{su}(3)$ generators as bilinears in the position and momentum operators (the angular momentum $\mathbf{L}$ and the quadrupole tensor $Q_{ij} = \frac{1}{2}(p_i p_j/m\omega + m\omega x_i x_j)$) and showing that they close into $\mathfrak{su}(3)$ under commutation. The converse — that $V(r) \propto r^2$ is the *only* central-field potential whose generators close into $\mathfrak{su}(3)$, so that the closure *forces* $V(r) \propto r^2$ — is the uniqueness sharpening established in the later dynamical-symmetry literature; we take the biconditional

$$V(r) = \tfrac{1}{2}m\omega^2 r^2 \iff \text{SU(3) dynamical symmetry}$$

in the form standard in the dynamical-symmetry literature [13, 14] (Iachello, Chapters 11–12 on spectrum-generating and degeneracy algebras). *Provenance caveat:* we have confirmed the forward direction (Jauch–Hill 1940) against its primary source, but not the *converse*—that su(3) closure forces $V \propto r^2$—against a specific numbered statement in either monograph. Theorem 2's converse half therefore rests on an imported result we flag as unverified at the pointer level.

Step 2 (Borel–Weil). The Borel–Weil theorem [12] identifies the symmetric irreducible representations $(N, 0)$ of SU(3) with the spaces of holomorphic sections of the line bundle $\mathcal{O}(N) \to \mathbb{CP}^2$. By restriction along the fibration $S^1 \to S^5 \to \mathbb{CP}^2$ (Eq. (8)), this is the same data as the homogeneous degree-N subspace $\mathcal{H}_N$ of the Hardy space $H^2(S^5)$ (Eq. (7)). Hence the $(N, 0)$ irreducible decomposition of an SU(3)-equivariant Hilbert space coincides exactly with the Hardy-space decomposition by holomorphic degree.

Step 3 (Bargmann–Segal–Hall). The Bargmann transform [7], made rigorous as a unitary intertwiner of group representations by Hall [9], establishes a unique (up to phase) unitary equivalence between $L^2(\mathbb{R}^3)$ with the Schrödinger representation of the harmonic oscillator and $F^2(\mathbb{C}^3)$ with the Bargmann representation, under which the harmonic oscillator Hamiltonian maps to $\hbar\omega(\hat{N}+3/2)$ (Eq. (5)). The Euler operator $\hat{N}$ is the unique self-adjoint operator on $F^2(\mathbb{C}^3)$ that is diagonal on the monomial basis with eigenvalue $a + b + c$ on $z_1^a z_2^b z_3^c$.

Combination. Suppose $V(r)$ is a central-field potential whose bound-state spectrum arises from the Euler operator on the Hardy-space sectors $\mathcal{H}_N$, with each $\mathcal{H}_N$ carrying the $(N, 0)$ irreducible representation of SU(3). By Step 2, the Hilbert space of $V(r)$ then decomposes under SU(3) into the symmetric irreducible representations $(N, 0)$, one per N, and the Hamiltonian is linear in $\hat{N}$ on each. By Step 3, this is exactly the image of the harmonic oscillator under the Bargmann transform. By the uniqueness of the Bargmann intertwiner (up to phase) and of the Schrödinger representation, the original Hamiltonian must be unitarily equivalent to the harmonic oscillator Hamiltonian. By Step 1, the only central-field potential with this property is $V(r) = \frac{1}{2}m\omega^2 r^2$, completing the converse. □ □

a. Remark on the converse step. The crucial input to the converse direction is the biconditional sharpening of Step 1 [13, 14]: the algebraic closure of the angular momentum and quadrupole generators into $\mathfrak{su}(3)$ *forces* $V \propto r^2$ (the forward implication is the original Jauch–Hill result [11]). This is the HO analog of the corresponding biconditional Fock–Bargmann result for the Coulomb problem (the Runge–Lenz closure forces $V \propto -1/r$). Both biconditional results are standard classical group theory [13, 14].

The harmonic oscillator and the Coulomb problem are therefore both rigid: the natural compact manifold and the natural operator together uniquely determine

the physical potential through the closure conditions of the dynamical symmetry algebra. These two rigidity results are independent (Coulomb uses S^3 and Laplace–Beltrami; HO uses S^5 Hardy space and the Euler operator) and do not derive from each other, but they have parallel structures and parallel proofs. Together they exhaust the two potentials that classical group theory identifies as having full SO(4) and SU(3) dynamical symmetries respectively.

A. Two-fermion entanglement rigidity — retracted

[RETRACTED 2026-08-22.] Earlier versions of this section carried a corollary of the HO rigidity theorem stating that *the spatial 1-RDM von Neumann entropy of the closed-shell two-fermion ground state on the Bargmann–Segal graph is identically zero for any central two-body interaction* $V(r_{12})$, on the mechanism that any central $V(r_{12})$ preserves the total HO quantum number $N_{\rm tot}$. **Both the claim and the mechanism are withdrawn.**

The mechanism is false. The Moshinsky–Talmi *bracket* conserves N — that is a property of the harmonic-oscillator coordinate transformation. The *potential matrix element* does not. A central $V(r_{\rm rel})$ is diagonal in the centre-of-mass quantum numbers and in $l_{\rm rel}$, but it *couples different relative-n*. Expanding the element, the bra bracket forces $2n + l + 2N_{CM} + \Lambda = N_{\rm bra}$ and the ket bracket forces $2n' + l + 2N_{CM} + \Lambda = N_{\rm ket}$, so $n - n' = (N_{\rm bra} - N_{\rm ket})/2$: the N-changing elements are exactly those with $n \neq n'$, and they are nonzero. For the Minnesota singlet at $b = 1$, $\langle n_{\rm rel}{=}0, l{=}0|V|n_{\rm rel}{=}1, l{=}0\rangle = +17.2$ MeV against a diagonal $\langle 0,0|V|0,0\rangle = -0.81$ MeV. (Magnitude caveat, 2026-08-29: this contrast is convention-sensitive — the b=1 diagonal sits near a zero crossing, and at the production convention ($b = 2.0364$, $\hbar\omega = 10$ MeV) the coupling is 0.48× the diagonal, *smaller* rather than larger. The load-bearing evidence for non-conservation is convention-independent: the measured commutator $\|[H_{HO}, V]\|_F/\|H_{HO}\|_F = 0.75$ and the nonzero corrected entropies below, not any single matrix-element ratio.)

The section's own contrast should have exposed this: it argued that the Coulomb case is nontrivial because $1/r_{12}$ "does not preserve any HO-like total-quanta quantum number", but $1/r_{12}$ *is* a central potential. Both statements cannot hold.

The zero was a code artifact. The published $S_{\rm full} = 0$ and $\|[H_{HO}, V]\|_F/\|H_{HO}\|_F < 10^{-15}$ were produced by an undocumented guard in the two-body matrix-element routine that returned zero whenever the bra and ket total quantum numbers differed, discarding the couplings above. With the guard removed the measured values are:

$N_{\rm max}$	E_0 (MeV)	$S_{\rm full}$	$\|[H_{HO}, V]\|_F/\|H_{HO}\|_F$
2	21.5437	0.0692	0.75
3	21.5182	0.0736	0.64
4	21.4363	0.0850	0.67

So the HO two-fermion ground state is not a single Slater determinant for a central two-body interaction, and H_{HO} and V do not commute. The π-freeness results of Section III B are unaffected — they concern the graph's eigenvalues and adjacency weights, not the interacting two-fermion state — but the claim that π-freeness extends "also to two-fermion entanglement" does not survive, and the asserted structural duality with Paper 27's nonzero Coulomb scaling is withdrawn with it: the two bases are not opposites on this axis.

Methodological note, recorded because it is the transferable part. The corroborating evidence was itself a symptom. The result was reported as robust partly because E_0 came out independent of $N_{\rm max}$ — but that independence was produced by the same guard, which froze the ground state inside a single $N_{\rm tot}$ block. Under the corrected computation E_0 decreases with basis size, as a variational calculation must. A quantity that comes out *exactly* zero, or *exactly* invariant, where the physics predicts only "small" or "weakly dependent", is a reason to ask what has been restricted.

B. Spinor-Lichnerowicz corollary: the operator-order discriminant

The HO rigidity theorem of Section V and the Coulomb/HO comparison of Section IV set a second-order Riemannian operator on S^3 (the Fock projection's $-\Delta_{\rm LB}$) beside the first-order, complex-analytic Bargmann–Segal Euler operator on $H^2(S^5)$. Both label sets are π-free, and π enters both sides through spectral sums at even argument (Section IV B). What the operator order governs is which zeta values appear: the first-order Euler ladder carries $\zeta(3)$ at $s = 4$ beside π^2 and π^4, while the S^3 Casimir ladder stays in $\mathbb{Q}[\pi^2]$.

The Dirac-on-S^3 Tier 3 sprint (Track T9) extends this result to spinor bundles. The squared Dirac operator on unit S^3 satisfies the Lichnerowicz identity $D^2 = \nabla^*\nabla + R/4$ [30], where $R = 6$. Its spectral zeta evaluates at every integer s to a two-term polynomial in π^2 with rational coefficients (Paper 18, Eq. (T9)), with *no odd-zeta content* at integer s (half-integer $s \geq 5/2$ carries odd zeta values, Paper 28).

The mechanism is algebraic: squaring maps the first-order Dirac sum $\sum g_n \cdot (\text{odd})^{-s}$ (which produces $\zeta_R(\text{odd})$ at odd s, as in Tier 1 Track D3) to the second-order sum $\sum g_n \cdot (\text{odd})^{-2s}$, which at integer s lies in $\mathbb{Q}\,\pi^{2s-2} + \mathbb{Q}\,\pi^{2s}$ by the Bernoulli formula (the quadratic degeneracy contributes two weights). At integer s the odd-zeta content of D is structurally lost upon squaring ($\zeta_{D^2}(5/2) = D(5)$ keeps it).

Corollary 1 (Operator-order transcendental discriminant)**.** *The operator-order distinction (first-order vs. second-order) is bundle-independent on round S^3 for the spectra n^2-1 and $(n+3/2)^2$ (partial fractions with rational poles). The second-order operators scalar Laplace–Beltrami and squared Dirac D^2 produce exclusively π^{even} spectral-zeta values at integer s (not every second-order operator: the spinor connection Laplacian $\nabla^*\nabla = D^2 - 3/2$, whose eigenvalues $(n+3/2)^2 - 3/2$ have irrational square roots, has $\zeta_{\nabla^*\nabla}(2) \notin \mathbb{Q}[\pi^2]$, Paper 28; an earlier version listed it) (half-integer s and spectral determinants are not covered). The Dirac operator D can produce odd-zeta content ($\zeta_R(3), \zeta_R(5), \ldots$) via half-integer eigenvalue sums. The bundle type (scalar vs. spinor) modulates rational coefficients and the eigenvalue/charge lattice (integer vs. half-integer) but does not change the transcendental class (the separation of the classes is conditional on the period conjecture, $\zeta_R(2k+1) \notin \mathbb{Q}(\pi)$).*

On round S^3 the second-order operators give only π^{even} at integer s, and this pattern persists on spinor bundles. The Bargmann graph is π-free because its entries are rational, as the S^3 labels are, not because its operator is first order. Cross-reference: Paper 18 §IV (taxonomy closure); Tier 1 Track D3 ($\zeta(3)$ from first-order Dirac); T9 debug memo (`debug/dirac_t9_memo.md`).

C. Gauge-structure corollary: reduced Wilson content on S^5

Paper 25 [31] observes that the GeoVac S^3 Coulomb graph at finite $n_{\max}$ is simultaneously a Fock discretization of S^3, a triangle-free simplicial 1-complex with discrete Hodge decomposition, and an abelian Wilson-type $U(1)$ lattice gauge structure with matter on nodes and a $U(1)$ connection on edges. The same three structural layers are available on the Bargmann–Segal graph of the present paper, but with reduced physical content. This corollary documents the transfer pattern.

a. Positive transfer (abelian $U(1)$ Wilson–Hodge structure). The Bargmann graph at $N_{\max} = 5$ has 56 nodes, 165 edges, one connected component, and Euler characteristic $V - E = -109$. The signed incidence matrix $B \in \mathbb{Z}^{56\times 165}$, oriented along the creation-operator direction ($N \to N+1$), gives a discrete exterior derivative $d_0 = B^\top$ with node Laplacian $L_0 = BB^\top$ and edge Laplacian $L_1 = B^\top B$. The first Betti number is $\beta_1 = E - V + c = 110$, counting the independent cycle classes that carry Wilson-loop holonomies in the abelian interpretation. Under a node-local $U(1)$ gauge transformation $\psi_v \mapsto e^{i\chi_v}\psi_v$, the dipole edge amplitudes $\langle w \,|\, z_q \,|\, v\rangle$ transform covariantly as $U_{vw} \mapsto e^{-i\chi_v} U_{vw} e^{i\chi_w}$. The combinatorial Hodge decomposition $\mathbb{R}^{165} = \operatorname{im} d_0 \oplus \ker L_1$ and the SVD identity between nonzero spectra of L_0 and L_1 both hold. At this level of abstraction, the Bargmann graph is a lattice gauge structure in exactly the Paper 25 sense.

b. Reduced content (no spectrum-computing role for L_0). An earlier version of this paragraph stated that on S^3 the node Laplacian $L_0 = D - A$ is conformally identified with the continuum [retracted 2026-09-25: graph-laplacian-converges-to-s3] S^3 Laplace–Beltrami operator, so that the Paper 25 Wilson dictionary (which identifies L_0 with the matter propagator's inverse) carried spectral content there. That premise is withdrawn (2026-09-25): L_0 on the (n,l,m) lattice preserves l, splits into $n_{\max}$ disconnected blocks and does not converge to Δ_{S^3}, whose exact action on the labels is the diagonal Casimir n^2-1 (Paper 7). The contrast drawn here therefore does not hold: on S^3 as on S^5, the node Laplacian is spectrally inert. On S^5 specifically, the Bargmann node Laplacian is spectrally inert for the physical HO spectrum: the HO energies $\hbar\omega(N+3/2)$ live in the diagonal (Eq. (17)), not in $L_0 = D - A$. The graph adjacency A encodes dipole transition amplitudes only, and Corollary 2 (item 1) shows that the node Laplacian L_0 computed from A plays no role in reconstructing the spectrum.

c. Reduced content (no Wilson SU(3) structure). The complex Hopf fibration $S^1 \to S^5 \to \mathbb{CP}^2$ of Eq. (8) suggests a potential non-abelian SU(3) gauge structure, with link variables valued in the fundamental rep **3** and node matter in the symmetric irreps $(N,0)$. This does not materialize. The dipole operator z_q is the fundamental **3** and intertwines $(N,0)\otimes(1,0) \to (N+1,0)$ as a Clebsch–Gordan coupling, not as a group element of SU(3). A Wilson lattice gauge structure requires the link variables to live in a fixed group (the same SU(3) on every edge) acting uniformly on a fixed matter irrep. The Bargmann graph has *different* SU(3) irreps on different shells (the representation dimension grows as $(N+1)(N+2)/2$), which is structurally a representation tower, not a lattice matter-gauge system. A pure-gauge SU(3) Wilson theory on the graph's edges is well defined; what fails is *natural* matter coupling, which needs a Clebsch–Gordan intertwiner on each link. The gauge group that couples naturally to the node wavefunctions is therefore $U(1)$, inherited from their phase ambiguity, not SU(3).

d. Reduced content (no $\mathbb{CP}^2$ base spectrum). The natural discrete Hopf quotient collapsing the m_l-fibers of the (SO(3)-adapted) Bargmann basis at fixed (N,l) gives a 12-sector graph at $N_{\max} = 5$:

$$(0,0),(1,1),(2,0),(2,2),(3,1),(3,3),$$
$$(4,0),(4,2),(4,4),(5,1),(5,3),(5,5)$$

with fiber dimensions $(2l+1) = 1,3,1,5,3,7,1,5,9,3,7,11$. The scalar Laplace–Beltrami spectrum on $\mathbb{CP}^2$ with the Fubini–Study Kähler metric of constant holomorphic sectional curvature $K = 4$ is [34]

$$\lambda_k = 4k(k+2), \qquad k = 0,1,2,\ldots, \tag{18}$$

with degeneracy $(k+1)^3$ (the dimension of the (k,k) SU(3) irrep). The 12-sector quotient Laplacian under the natural multiplicity weighting has eigenvalues

$\{0, 2.22, 4.87, 4.94, 11.65, 12.11, 18.74, 28.09, 33.58, 50.61, 63.77, 109.43\}$. The best single-scale rescaling against the analytic CP^2 sequence $\{0, 12, 32, 60, 96, 140, \ldots\}$ has a maximum relative residual of 24.98%; the ratios $\lambda_{\mathrm{quot},k}/\lambda_{\mathrm{CP}^2,k}$ vary between 0.08 and 0.19 across the 12 pairs, with no uniform rescaling. The m_l-quotient is therefore a combinatorial SO(3)-labeled graph on 12 sectors, not a spectral discretization of $\mathbb{CP}^2$. At this $N_{\max}$, there are simply too few sectors to represent the four-dimensional $\mathbb{CP}^2$ Laplacian, and the weighting scheme in the quotient is inherited from SO(3) branching, not from the Kähler volume form on $\mathbb{CP}^2$.

Corollary 2 (Gauge-structure reduction on the Bargmann graph)**.** *The abelian $U(1)$ Wilson–Hodge structure of the S^3 Coulomb graph (Paper 25) transfers to the Bargmann–Segal S^5 graph at the combinatorial level (incidence matrix, node/edge Laplacians, discrete Hodge decomposition, β_1 Wilson-loop classes), but the physical content of the structure is strictly reduced:*

1. *The node Laplacian $L_0 = D - A$ is spectrally inert on S^5 (the HO spectrum is in the diagonal).* Withdrawn contrast (2026-09-25): *this item formerly said L_0 carries the Coulomb spectrum on S^3 through a conformal identification with the continuum Laplace–Beltrami operator; it does not (L_0 does not converge to Δ_{S^3}, whose action on the labels is the diagonal Casimir, Paper 7), so this item is no longer a reduction relative to S^3.*

2. *The gauge group that couples naturally to the node matter is $U(1)$. Pure-gauge* SU(3) *Wilson theory on the edges is well defined, but no natural* SU(3) *gauge–matter structure emerges from the $(N, 0)$ symmetric irreps, because transitions between distinct irreps are Clebsch–Gordan intertwiners, not group elements.*

3. *The m_l-fiber-collapse quotient does not produce a spectral discretization of $\mathbb{CP}^2$ at finite $N_{\max}$.*

The corollary extends the Coulomb/HO asymmetry of Section IV by adding gauge-structure content to the third layer catalogued in §IV D (Wilson gauge with natural matter coupling: on S^3 the links and the spin-1/2 matter share a representation, an argument not yet realised as a coupled construction, Paper 30 § open; on S^5 pure-gauge SU(3) is well-defined but natural $(N, 0)$ matter needs a Clebsch–Gordan intertwiner). This layer rests on representation theory, not on L_0, and is its own layer; the "spectrum" layer ("spectrum: yes S^3, no S^5") and the calibration-π layer are withdrawn (2026-09-25, 2026-10-19). The universal content — that the Wilson–Hodge vocabulary applies combinatorially to any simplicial 1-complex — is preserved, but the Coulomb-specific content — a matter-gauge double in which matter and links share one representation (argued on S^3, not yet constructed) — is not. (An earlier version also listed, as Coulomb-specific, that the graph Laplacian carries the physical spectrum through a conformal identification with the continuum Laplacian; it does not, on either sphere.)

A practical consequence for the Paper 2 fine-structure-constant Observation [22]: the S^5 analog of the Paper 2 combination rule $K = \pi(B + F - \Delta)$ does not exist in natural form. A candidate F-analog (the Dirichlet series $\sum_{N\geq 1} g_N^{\mathrm{HO}}/N^{d_{\max}}$ with $g_N^{\mathrm{HO}} = (N+1)(N+2)/2$ and packing exponent $d_{\max} = \dim \mathbb{C}^3 = 6$) evaluates to $\frac{1}{2}[\zeta_R(4) + 3\zeta_R(5) + 2\zeta_R(6)]$, a mixed even/odd zeta combination with no clean $\zeta_R(2) = \pi^2/6$ projection; and Paper 23's Fock projection rigidity theorem establishes that the S^3 conformal projection is unique to $-Z/r$ and has no S^5 counterpart, so the B and Δ analogs (both tied to that projection) also do not lift. The S^5 graph is thus a lattice gauge structure with abelian $U(1)$ content but without α-like spectral content, consistent with the operator-order transcendental discriminant of Section V B. Cross-reference: Paper 25 §VII.1 (the open question answered here); debug memo `debug/s5_gauge_structure_memo.md`; numerical data `debug/data/s5_graph_spectrum.json`.

VI. IMPLICATIONS

A. For the fine-structure constant Observation

Paper 2 [22] observes that the fine-structure constant approximately satisfies (to 8.8×10^{-8} relative; an Observation) the cubic equation $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B + F - \Delta)$, where $B = 42$, $F = \zeta(2) = \pi^2/6$, $\Delta = 1/40$ are spectral invariants of the Hopf fibration $S^1 \to S^3 \to S^2$. The plain sum $K = 137.036064$ matches $1/\alpha$ to 4.8×10^{-7}; the 8.8×10^{-8} agreement is that of the cubic's root (Paper 2's self-consistency condition). In a combinatorial search the formula ranks among the top ~ 10 of 1.92×10^9 formulas ($p = 5.2 \times 10^{-9}$ per draw), a statistic that does not account for the unregistered search that produced it. The combination rule $K = \pi(B + F - \Delta)$ is an empirical Observation.

The Coulomb / HO comparison of Section IV gives no structural support to the prefactor π: both sides carry π in a measure and in spectral sums (Proposition 1), so the presence of a π in K does not by itself point to the Coulomb problem. On the corpus's accounting the leading π is the M_1 volume content of S^3 (Papers 31, 55), a reading at Paper 18's tier. Paper 2's combination rule remains an Observation; its status is unchanged by this paper.

B. For nuclear quantum simulation

Paper 23 [27] constructs nuclear shell-model qubit Hamiltonians on the harmonic oscillator single-particle basis, producing the deuteron and helium-4 qubit Hamiltonians via the Minnesota nucleon-nucleon potential and

Moshinsky–Talmi brackets. Theorem 1 of the present paper implies that the single-particle structure of these Hamiltonians is exactly rational at the graph level: the diagonal HO energies, the orbital labeling, and the dipole transition selection rules are all built from rational arithmetic on integers. Any floating-point error in nuclear Pauli coefficients comes from the nucleon–nucleon interaction matrix elements (which involve Gaussian integrals via Moshinsky–Talmi brackets), not from the single-particle basis or the graph topology. This is a precision guarantee for nuclear qubit Hamiltonians built within the GeoVac framework.

C. The universal / Coulomb-specific partition completed

Combined with Paper 22 [26] (potential-independent angular sparsity) and Paper 23 [27] (Fock projection rigidity theorem), the present paper completes a three-way partition of the GeoVac framework's claims:

a. Universal across spherical fermion systems. The angular-momentum selection rules from Wigner $3j$ symbols and the Gaunt integral; the ERI density bound (at $l_{\max} = 3$: 6.06% under the full-Gaunt global-M_L rule, 1.44% under the stricter pair-diagonal restriction—independent of $V(r)$ either way); the block-diagonal structure of two-body operators; and the exactly Q-linear composed Pauli count, $N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis (the retired pair-diagonal rule gave $O(Q^{2.5})$). These are the Paper 22 results.

b. Coulomb-specific. The S^3 Fock conformal projection; the Hopf bundle structure underlying the α Observation; the matched scale $\kappa = -1/16$ (it matches the graph band edge to $-\frac{1}{2}$; its coincidence with the Fock-Jacobian $1/16$ is an Observation). These are the Paper 23 / Track NH results and the present paper.

c. Harmonic-oscillator-specific (this paper). The S^5 Hardy-space discretization; the linear-affine projection $E_N = \hbar\omega(N + 3/2)$; the bit-exactly π-free graph at every $N_{\max}$; the embedding π^{-3} in the continuous Bargmann measure that the discrete graph never invokes; the Bargmann–Segal rigidity theorem (Theorem 2).

The two natural geometries (S^3 for Coulomb and S^5 Hardy for HO) are not in competition; they are the discrete encodings of two physically distinct potentials with two distinct dynamical symmetry groups. The asymmetry between them is structural and informs which features of the GeoVac framework extend to nuclear, atomic, and molecular systems beyond the electronic Coulomb problem.

VII. CONCLUSION

We constructed a discrete graph encoding of the three-dimensional isotropic harmonic oscillator on the holomorphic sector of S^5, proved that the graph and its projection to physical energies are bit-exactly π-free in exact rational arithmetic (Theorem 1), and showed in the Kustaanheimo–Stiefel frame that the placement of π does not separate it from the Coulomb problem (Proposition 1). We stated and proved the harmonic oscillator analog of the Fock projection rigidity theorem (Theorem 2), conditional on the imported SU(3)-closure converse, based on the biconditional uniqueness of SU(3) dynamical symmetry (forward direction due to Jauch–Hill; the converse, attributed to Wybourne and Iachello, is imported rather than proved here), the Borel–Weil identification of $(N, 0)$ irreducibles with the holomorphic Hardy-space sectors, and the Bargmann–Segal–Hall unitary intertwiner. The same oscillator is reached from the finite label set of a spin by the large-spin contraction, which keeps the label n, releases only the cap $n \leq 2S$, and lets in $\pi^{1/2}$ and e^{-x^2} only as limits of rational label data (Section III C).

In the Kustaanheimo–Stiefel frame the Coulomb problem is itself a constrained four-dimensional oscillator: its labels, degeneracy and Casimir sit on an integer oscillator lattice, its projection is linear at fixed energy, and the lift costs one fibre volume $\pi/4$ that cancels in every ratio. Both sides carry π in a measure and in spectral sums at even argument. The Coulomb/HO asymmetry is carried by representation theory and arithmetic (Layers 3–7 of Section IV D), not by where π enters.

Future work includes extending the discretization to anisotropic harmonic oscillators (which break SU(3) to a smaller subgroup, with corresponding modifications of the manifold), and identifying the natural compact manifolds for other dynamical-symmetry-classified central potentials such as the Morse oscillator (SU(1, 1)), the Kepler problem in n dimensions (SO($n+1$)), and the Pöschl–Teller hierarchy.

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